

Classification of Solid -

1) On the basis of solid structure -

- a) crystalline
- b) Amorphous
- c) semicrystalline

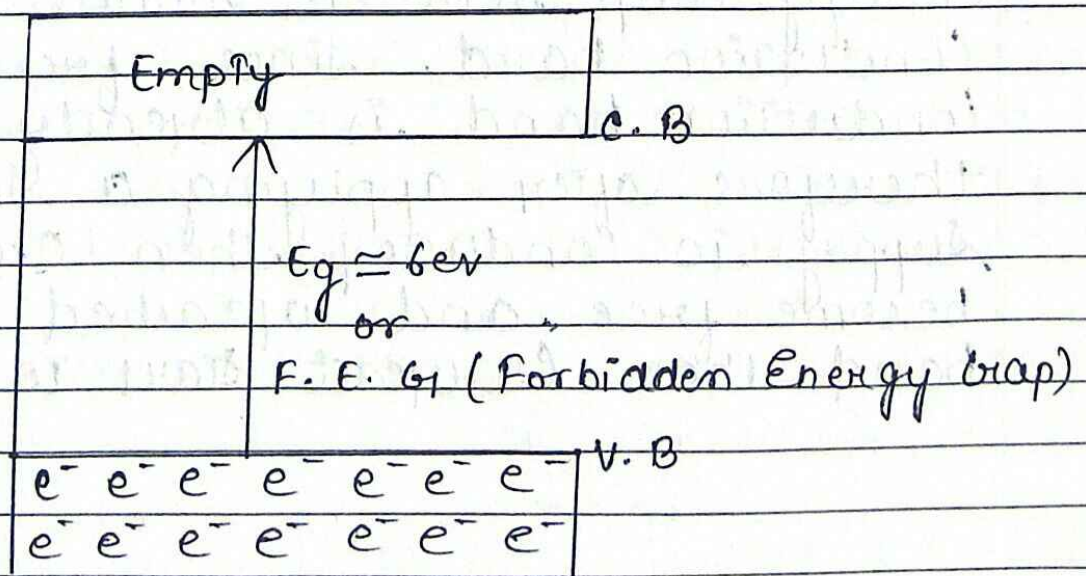
2) On the basis of Conductivity / Band Gap -

- a) conductor
- b) Insulator
- c) Semiconductor

* Insulator :-

In the Term of Energy Band, the Valance band is full with electron while conduction band remains empty and forbidden energy gap will very high $\approx 6\text{eV}$.

Then electron cannot be reached to conduction band after supply the Energy. Hence there is no free e^- available in conduction band under Normal Condition.



NOTE-

we use Semiconductor instead of Conductor because in semiconductor, there is abundance amount of e^- is present. when e^- starts moving it produces heat flow, which is very dangerous or it is not suitable for use.

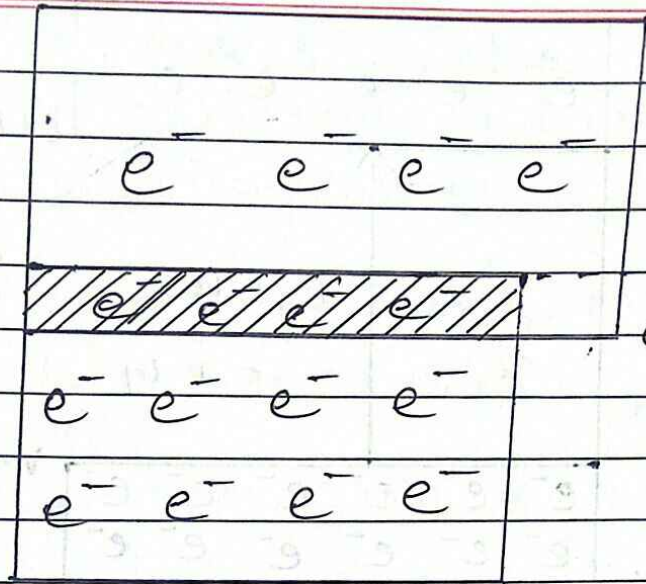
- Insulator has Negative Temperature Coefficient -

- +ve Temperature $\rightarrow$ Temp $\uparrow$, Resistivity $\uparrow$
Coefficient

- -ve Temperature $\rightarrow$ Temp $\uparrow$ Resistivity $\downarrow$
Coefficient

* Conductor -

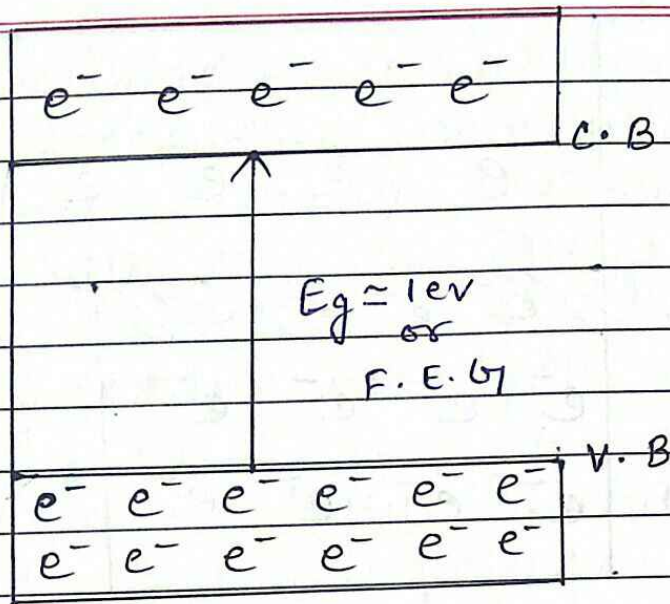
Conduction Band and valance band overlap to each other, Hence large no. of free e^- already present in conduction Band. There is no forbidden energy gap between valance band and conduction band. Since, free e^- in the conduction band is already present therefore after applying a small energy supply to conductor then electron become free and reached to conduction band then current start to flow.



- Conductor has Positive Temperature Coefficient.
- In Conductor, Current flow due to electron.

* Semiconductor $\rightarrow$ It is a substance which has few free e^- at room temperature.
Ex. - Si, Ge etc.

At very low temperature, conduction band will be empty and forbidden energy gap will be small approximately (1 eV) and almost filled valance band. When Energy supply to semiconductor then e^- reached to semiconductor conduction band and current starts to flow.



$$E_g \Rightarrow 1.1 \text{ eV (Si)}$$

$$E_g \Rightarrow 0.7 \text{ eV (Ge)}$$

In Semiconductor, current conduction take place due to holes and electrons. Semiconductor has negative temperature coefficient. Covalent bond present in Semiconductor.

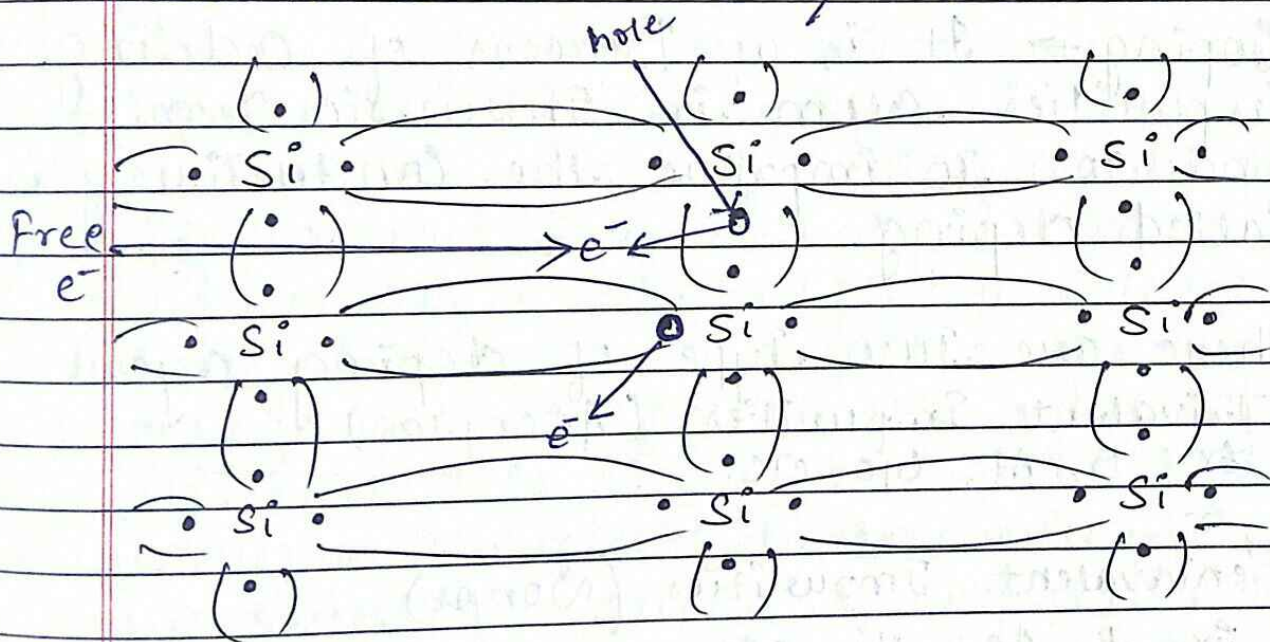
There are two types of Semiconductor-

- ① Intrinsic Semiconductor
- ② Extrinsic Semiconductor

* Intrinsic Semiconductor-

- A pure semiconductor is called Intrinsic Semiconductor.
- A Semiconductor which is free from any impurity.

- The no. of free e^- (n_e) will be equal to no. of holes (n_h) since resistance decrease in increase in Temperature.
- Mobility of no. of e^- is greater than no. of holes.
- Intrinsic Semiconductor is electrically neutral.
- It has very poor conductivity at room Temperature.
- In Intrinsic Semiconductor, current flow due to both electron & hole.



* Extrinsic Semiconductor -

1) when some suitable impurities are doped in Intrinsic Semiconductor then Extrinsic semiconductor will be formed.

2) Generally, Conductivity of Pure Semiconductor will be very low so for increase the conductivity some external material should be mixed with semiconductor then the process of mixing of impurities is called doping and semiconductor is called doped semiconductor or Extrinsic Semiconductor.

Doping $\rightarrow$ It is a process of adding impurities atom in Intrinsic Semiconductor to improve the conductivity is called doping.

• There are two types of doping agents -

$\rightarrow$ Trivalent Impurities (Acceptors)
Ex - B, Al, Ga etc.

$\rightarrow$ Pentavalent Impurities (Donor)
Ex - P, As, Bi etc.

On the basis of doping, there are two types of Extrinsic Semiconductor -

1) N - Type Semiconductor

2) P - Type Semiconductor

NOTE

Doping Profile - $10^6 : 1$

N-Type Semiconductor $\rightarrow$ When Pentavalent element doped with Pure Semiconductor. then N-Type Semiconductor obtained.

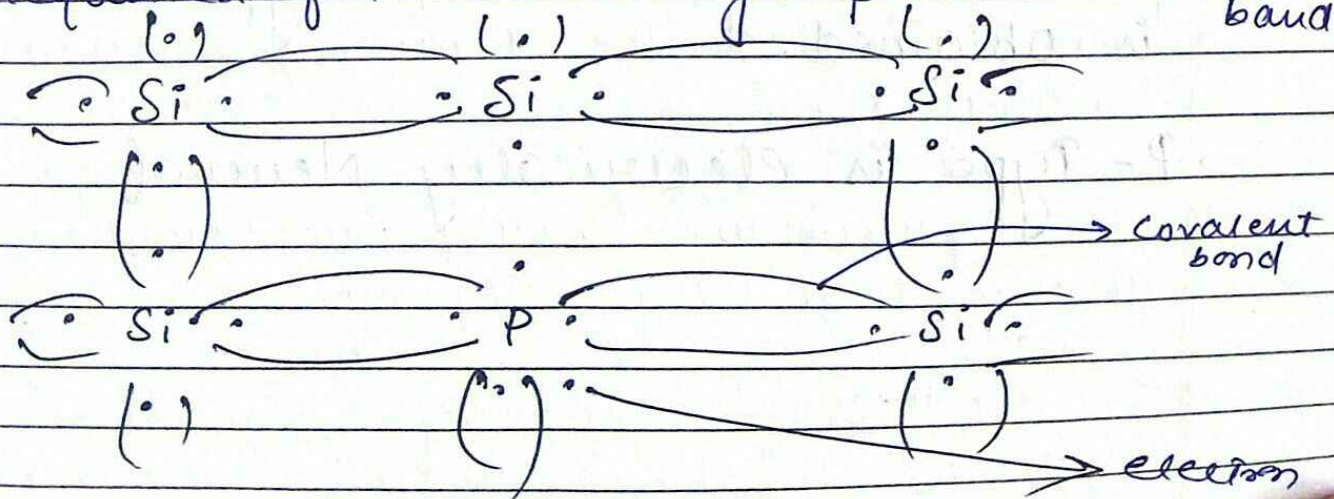
$\rightarrow$ N-Type Semiconductor is electrically Neutral.

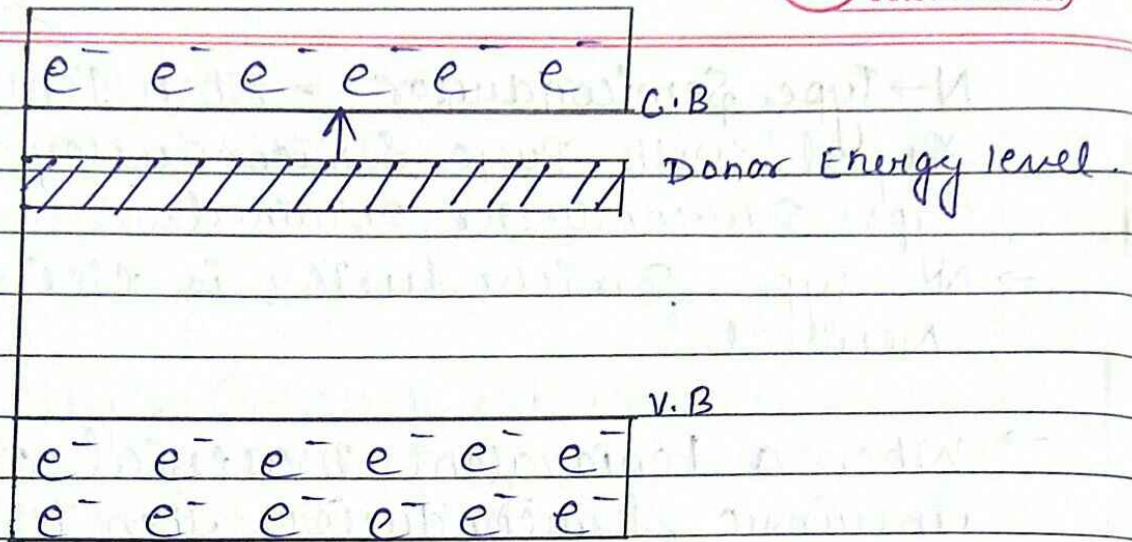
$\rightarrow$ When a Pentavalent material added to intrinsic semiconductor then its four e^- from covalent bond with surrounding four silicon atom and 5th valance e^- of Pentavalent atom find no place in covalent bond and it free to move in crystal.

$\rightarrow$ These 5th e^- is responsible for conduction of current in N-Type Semiconductor.

On the basis of Energy Band Diagram when donor impurity are added to pure semiconductor, the donor atom generates discrete energy level which is just below the conduction band.

Therefore, very small amount of Energy is required for an e^- to jump in the conduction band.





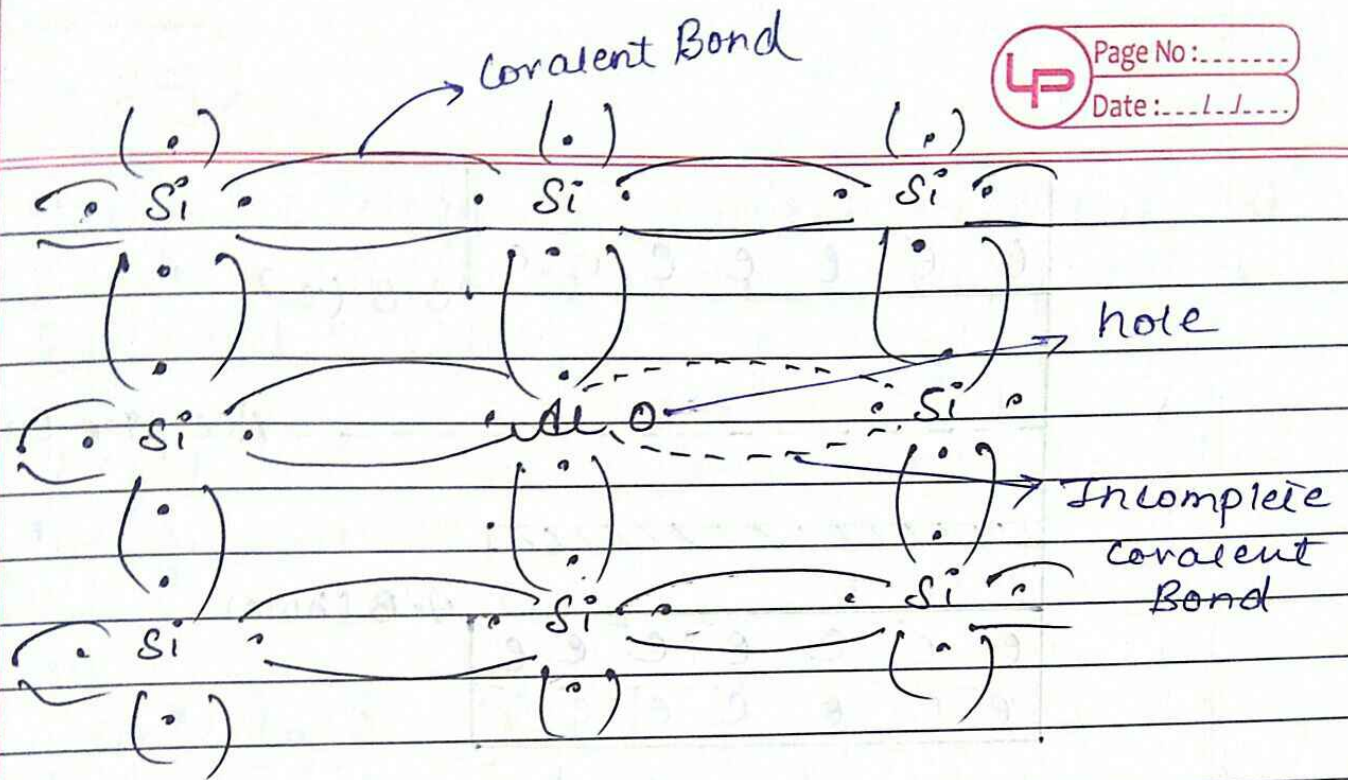
In N-Type thermal Energy at Room Temperature still generates few e^- hole pair.

In N-Type Semiconductor, the no. of e^- is more than No. of hole. Therefore, electron is majority charge carrier and hole is minority charge carrier.

P-Type Semiconductor -

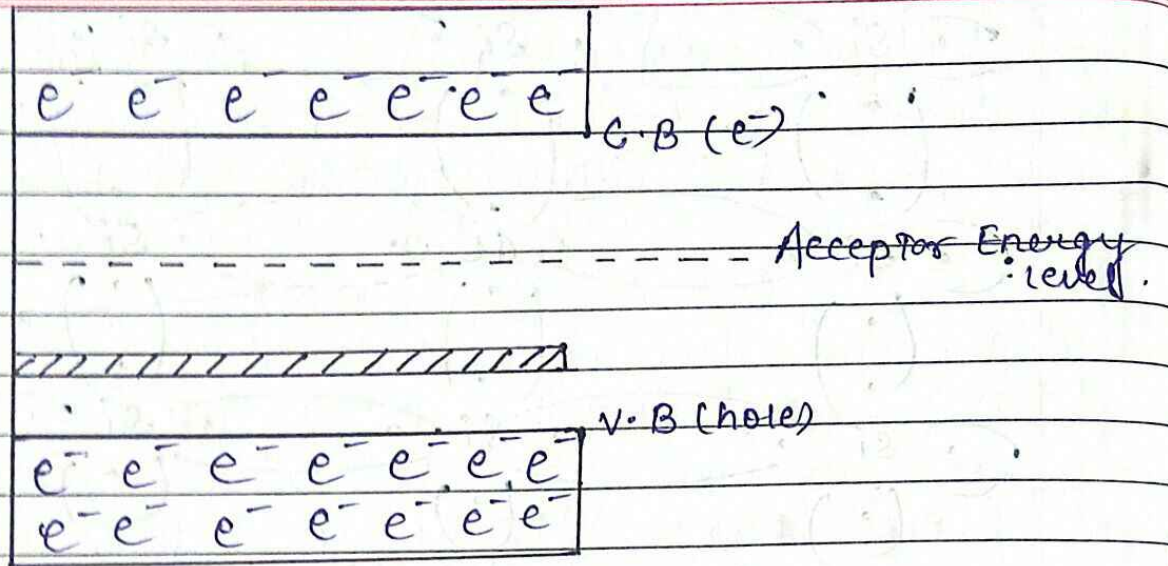
When Tri-valent element doped with Semiconductor then P-Type Semiconductor is obtained.

→ P-Type is electrically Neutral.



In P-type Semiconductor the three valence e^- of the impurity (Trivalent) atom will form the Covalent Bond with three Si atom. But one covalent bond left incomplete and give one extra hole. These hole in P-type Semiconductor is responsible for electric conduction.

On the basis of Energy band diagram when acceptor impurities (Trivalent) are added to pure semiconductor then it produce distinct energy level which is just above the valance band. Therefore, very small amount of Energy is required for an e^- to leave the valance band which is responsible for current conduction in P-type semiconductor.



In P-type Semiconductor Thermal Energy at Room Temperature still generate a few electron hole pair.

In P-type Semiconductor, No. of hole is more than No. of e^- . Hence, Hole ^{are} Majority charge carrier whereas e^- are minority charge carrier.

Equilibrium Carrier Concentration

- The No. of Carrier in the Conduction band and V.B with No externally applied bias is called the Equilibrium carrier Concentration.
- The Equilibrium Carrier Concentration can be increased through doping.
- Semiconductor contain Majority & Minority Carrier, The More abundant charge carrier

are the Majority carrier and the less abundant charge carrier is Minority Carrier.

→ The Product of Majority and Minority charge carrier is constant.

MASS ACTION LAW for Intrinsic

The law of Mass Action states that the Product of No. of electron in Conduction Band and the No. of hole in Valance band is constant at a fixed Temperature and it is independent of amount of donor and acceptor impurities added.

$$n \cdot p \Rightarrow n_i^2 \Rightarrow \text{Constant}$$

→ Mathematical Expression

where,

n - No. of e^- in Conduction Band.

p - No. of hole in Valance Band.

n_i - Intrinsic carrier Concentration

G_p - Generation of hole

R - Recombination of e^- & hole

At Equilibrium, $G \Rightarrow R$

Means Rate of Generation $\Rightarrow$ Rate of Recombination

$$R \propto n$$

$$R \propto p$$

$$R \propto np$$

$$\boxed{R \Rightarrow \tau \cdot np} \quad \text{--- (1)}$$

Now,

$$G \propto n_i^2$$

$$\boxed{G \Rightarrow \tau n_i^2} \quad \text{--- (2)}$$

Since, from Eq - (1) & (2) :-

$$R \Rightarrow G$$

$$\cancel{\tau \cdot n \cdot p} \Rightarrow \cancel{\tau n_i^2}$$

$$\boxed{n \cdot p \Rightarrow n_i^2}$$

$\rightarrow$ For Intrinsic Semiconductor

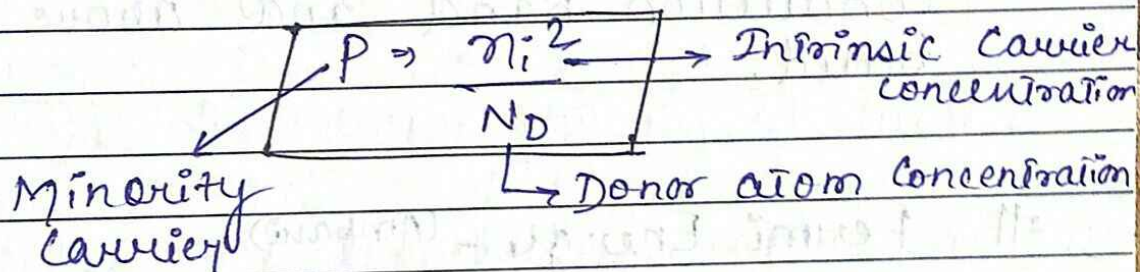
LAW OF MASS ACTION

for Extrinsic Semiconductor

1) For N-Type, $n \approx N_D$

$$n \cdot p \approx n_i^2$$

$$N_D \cdot p \approx n_i^2$$

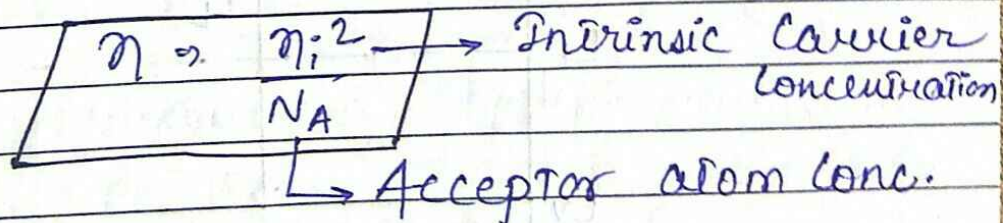


2) For P-Type,

$$p \approx N_A$$

$$n \cdot p \approx n_i^2$$

$$n \cdot N_A \approx n_i^2$$



Above shows that No. of Minority Carrier decreases with increase in Doping level.

(Prime Minister)

Fermi-level -

→ The highest Energy level that e^- can occupy at absolute zero temperature ($0K$, $-273^\circ C$).

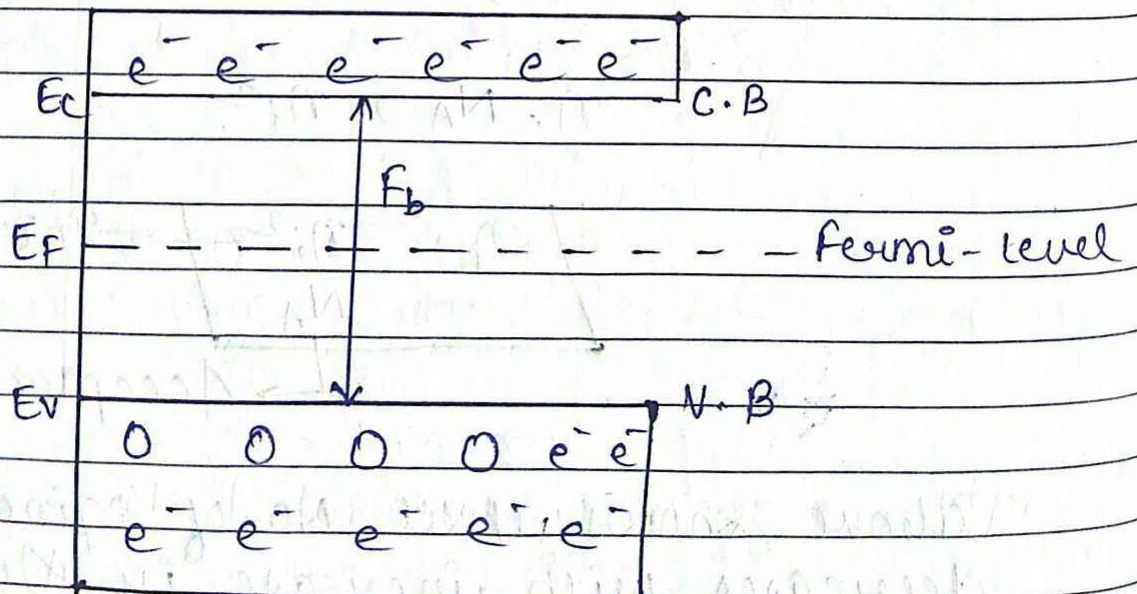
The fermi level lies between valance band and conduction band.

→ The Fermi level always exist below the conduction band and above the valance band.

Fermi Energy - (Ambani)

→ It is a Maximum Energy Possessed by free e^- at absolute zero temperature

Fermi-level of Intrinsic Semiconductor -



$E_c \rightarrow$ Energy of Conduction Band

$E_f \rightarrow$ Energy of Fermi-level

$E_v \rightarrow$ Energy of Valance Band.

* In Intrinsic Semiconductor, no. of e^- at C.B $\Rightarrow$ no. of holes in V.B.

No. of e^- in Conduction Band -

$$n_e \Rightarrow N_c \cdot e^{-\left(\frac{E_c - E_f}{kT}\right)}$$

No. of holes in Valance Band -

$$n_h \Rightarrow N_v \cdot e^{-\left(\frac{E_f - E_v}{kT}\right)}$$

n_e and $n_h \Rightarrow$ No. of e^- in Conduction Band and No. of hole in Valance Band.

N_c & $N_v \rightarrow e^-$ density in C.B and hole density in V.B respectively.

In Intrinsic Semiconductor $N_c \Rightarrow N_v$.

$$\therefore N_c \cdot e^{-\left(\frac{E_c - E_f}{kT}\right)} \Rightarrow N_v \cdot e^{-\left(\frac{E_f - E_v}{kT}\right)}$$

$$e^{-\left(\frac{E_c - E_f}{kT}\right)} \Rightarrow e^{-\left(\frac{E_f - E_v}{kT}\right)}$$

$$\frac{E_C - E_F}{kT} \rightarrow \frac{E_F - E_V}{kT}$$

$$E_C - E_F \rightarrow E_F - E_V$$

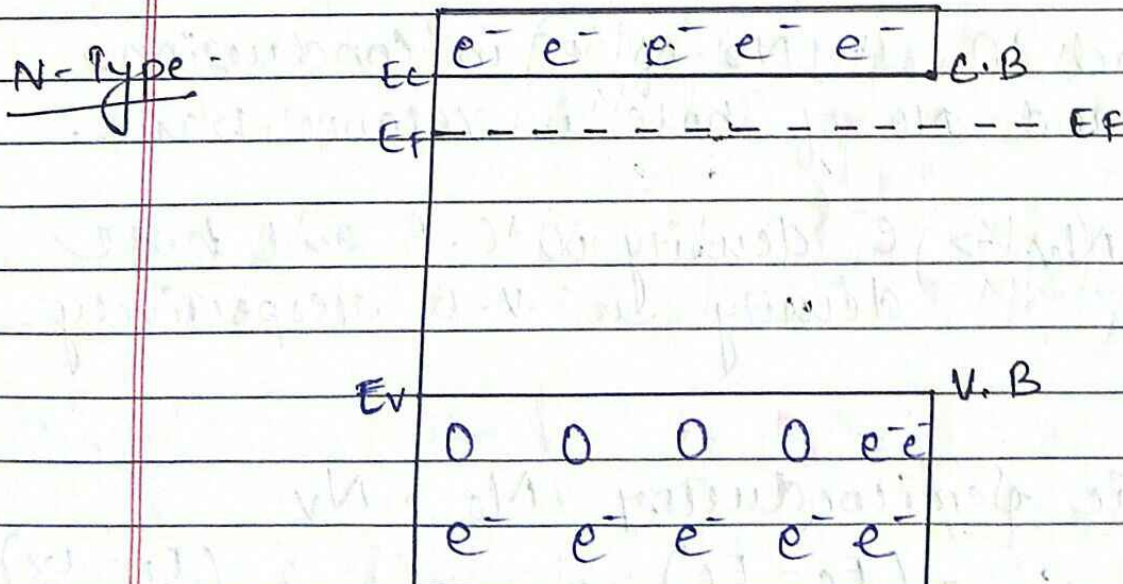
$$E_C + E_V \rightarrow 2E_F$$

$$E_F \rightarrow \frac{E_C + E_V}{2}$$

$T \rightarrow$ Temperature in Kelvin

$k \rightarrow$ Boltzmann constant.

Fermi - level of Extrinsic Semiconductor



$N_e = N_D$ (concentration of $e^- \Rightarrow$ donor atom).

$$\Rightarrow N_e \Rightarrow N_c \cdot e^{-\left(\frac{E_c - E_f}{kT}\right)}$$

$$\Rightarrow N_D \Rightarrow N_c \cdot e^{-\left(\frac{E_c - E_f}{kT}\right)}$$

$$\Rightarrow \frac{N_c}{N_D} \Rightarrow e^{\left(\frac{E_c - E_f}{kT}\right)}$$

$$\Rightarrow \ln \left(\frac{N_c}{N_D} \right) \Rightarrow \ln e^{\left(\frac{E_c - E_f}{kT}\right)}$$

$$\Rightarrow kT \ln \left(\frac{N_c}{N_D} \right) \Rightarrow E_c - E_f$$

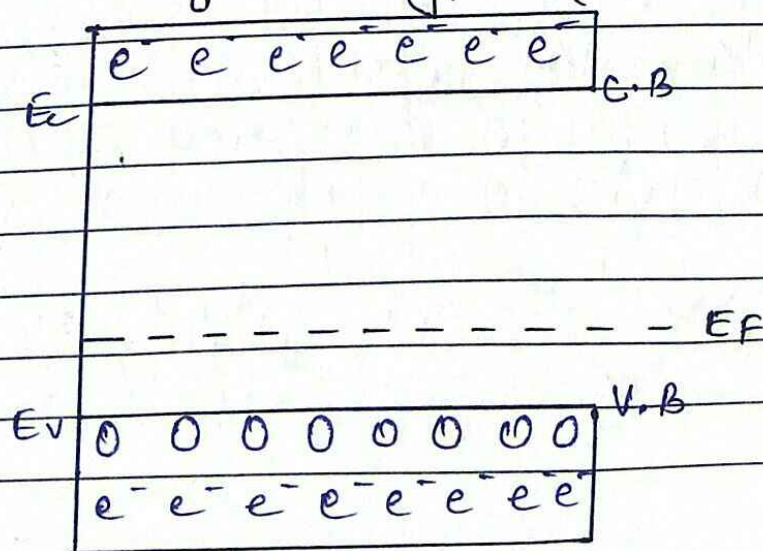
$$\therefore E_f \Rightarrow E_c - kT \ln \left(\frac{N_c}{N_D} \right)$$

Fermi level is depend on :-

- ① Temperature
- ② Doping

Fermi level of P-Type Semiconductor-

P-Type



$$p \approx N_A$$

$$N_h \approx N_v \cdot e^{-\left(\frac{E_F - E_v}{kT}\right)}$$

$$N_h \approx N_A$$

$$N_A \approx N_v \cdot e^{-\left(\frac{E_F - E_v}{kT}\right)}$$

$$\frac{N_v}{N_A} \approx e^{\frac{E_F - E_v}{kT}}$$

$$\ln\left(\frac{N_v}{N_A}\right) \approx \ln e^{\frac{E_F - E_v}{kT}}$$

$$\ln\left(\frac{N_v}{N_A}\right) \approx \frac{E_F - E_v}{kT}$$

$$kT \ln\left(\frac{N_v}{N_A}\right) \approx E_F - E_v$$

$$E_F \approx E_v + kT \ln\left(\frac{N_v}{N_h}\right)$$